WINDOWS OS

- Windows is a Graphical user Interface operating System and it is a System S/W.
- windows os is developed and Published by Microsoft.
- windows os was released on NoverHber 20, 1985.
- Microsoft windows is user friendly Popular and Most used operating System.
- First version of windows, Microsoft windows 1.0 was relased on November 20. 1985.

windows 1.0 - is a 16-bit graphical (OS) environHent ~~relased~~ released on November 20, 1985.

windows 2.0 :- released after Two years of version 1.0 (1987)
- First windows version of Ms word and Excel.

windows 2.1 :- DeceMber 28, 1987 - May 27, 1988.
→ Less than a year after the relase of windows/286. 2.1 and windows/386 2.1 were released on May 27 1988.

- windows 3.0 :- Third Major release of MS windows.
→ First Successful OS.
→ A powerful rival to Apple mac and the commodore Amigaon the GUI front.

windows 3.1 :- The OS that superseded windows 3.0.
→ ~~The~~ only an update to the First windows OS under version 3.0
→ April 1992.
→ First windows OS user windows Logo (Flying windows Logo).

workgroup 3.1
↳ October 1992

windows NT 3.1
December 1993

windows NT Workstation 3.5 :- Second release of the Microsoft windows NT OS.

→ used for Workstation / client computers.
→ September 21, 1994.
→ Goals - to increase the speed of the OS. as a result, the project was given the codename "Daytona" in reference to Daytona International Speedway.

windows NT 3.51

- Third release of Ms windows NT line of OS.
- "Cairo".
- May 30, 1995.
- nine (9) Months after 3.5.

windows 95 (windows 4.0)

- International Codename "Chicago" (Illinois).
- Consumer- ~~ang~~ oriented GUI-based OS.
- August 24, 1995.
- Fourth Major release.

Feature / Aspect	windows	Linux	Unix
1) Developer	Microsoft	community (open source)	AT&T Bell Labs
2) Source code	closed-source	open-source	Mostly closed source.
3) Cost	Paid (license needed)	Paid (for Most versions)	Free (Mostly)
4) users	General users, gamers	Enterprises servers	Developers, Sysadmins, servers.
5) security	Moderate	Strong	very strong
6) customi-zation	very limited	Limited	Highly customizable.
7) used in	Personal PCs, offices.	servers, universities.	servers, desktops, embedded systems.

* **windows:-**
- Developer: Microsoft
- Source: closed source (Proprietary)
- user interface: Graphical by default (GUI - heavy)
- Usage:- commonly used in personal computers, offices, gaming
- File system: NTFS, FAT32.
- command Line: CMD/Powershell (less powerful compared to unix/Linux shells)
- Software compatibility:- compatible with most mainstream software, especially for gaming and office work.
- Security:- More vulnerable to Malware due to widespread use, though improving.

* **Linux:-**
- Developer: community-based (originally by Linus Torvalds)
- Source: open-source (Free to use and modify)
- user interface:- Usually CLI-based, but supports GUI (GNOME, KDE) etc.
- Usage:- widdely used in servers, developers, embedded systems, cybersecurity.
- File system: ext4, ext3, Btrfs, XFS.
- command Line: Bash, zsh, etc.
- security:- Generally considered very secure due to unix-like permissions and open-source nature.
- Customization: Highly customizable.

Unix: - • Developer :- originally AT&T Bell Labs.

• Source: - Mostly closed Source (Commercial)

• User Interface: - CLI-based, some GUI environments.

• Usage :- used in enterprise servers, mainframes, academic and industrial systems.

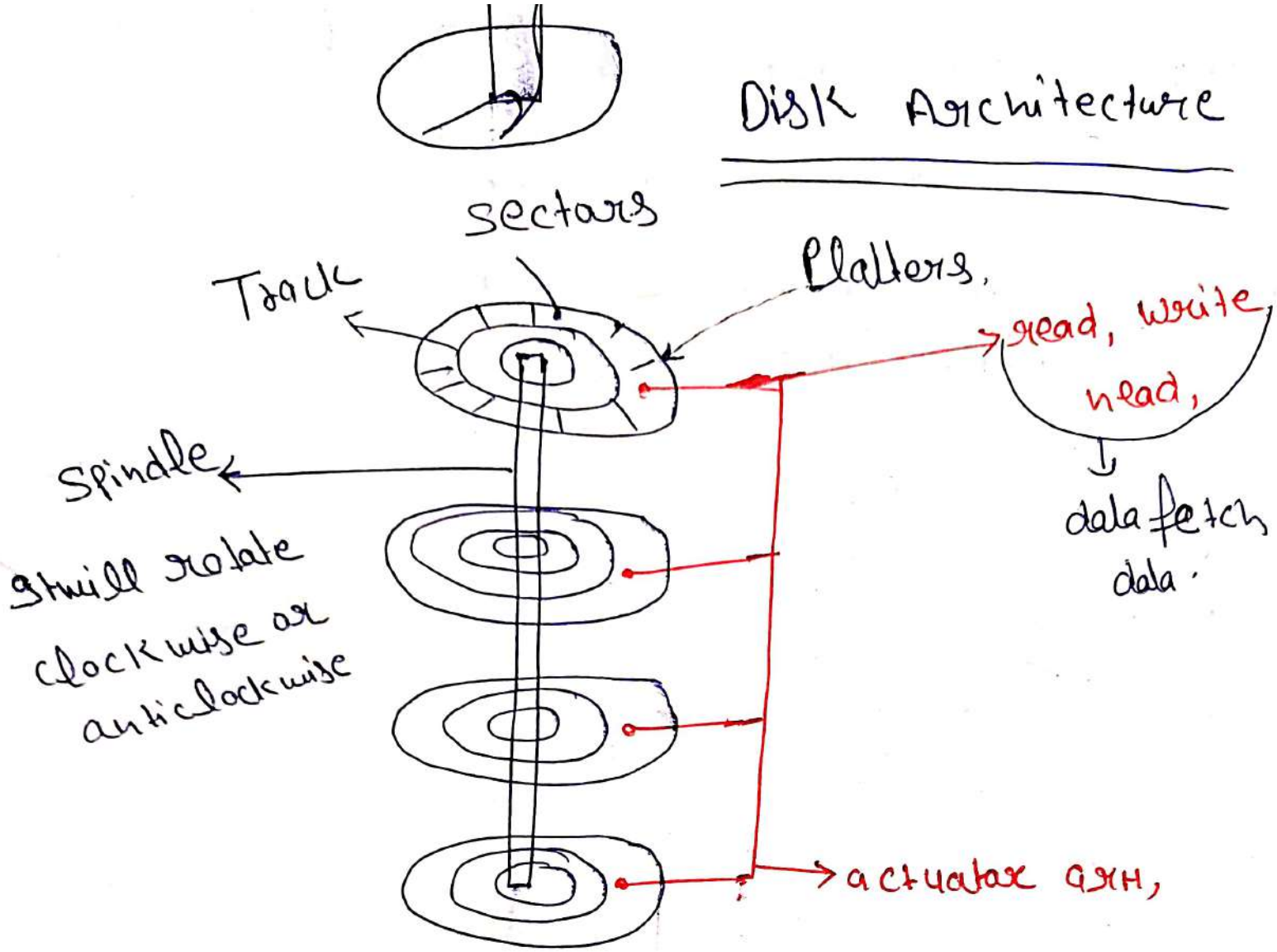
File System :- UFS, ZFS.

Command Line: - Bourne shell, C Shell, Korn shell, etc.

Example: - Solaris, AIX, HP-UX.

Legacy: - unix is the base philosophy for both Linux and MacOS.

Disk Architecture



Disk Size = Platters → Surface → Track → Sectors → Data.

Disk Access Time

1) **Seek (T):** - Time Taken by R/W head to reach desired Track.

2) **Rotation (T):** - Time Taken for one full Rotation (360°)

3) **Rotational latency:** - Time Taken to desired sector. (half of Rotation Time)

4) **Transfer:** -

$$\frac{\text{Data to be Transfer}}{\text{Transfer Rate}} \quad \text{~~Transfer Rate~~}$$

~~Transfer rate: (Data Rate:) { No of Head × Capacity of one Track ×~~

$$\frac{\text{Data to be Transfer}}{\text{Transfer Rate}}, \quad \text{Transfer rate (Data rate)} \left\{ \begin{array}{l} \text{No. of Heads} \\ \text{Surface} \end{array} \times \begin{array}{l} \text{Capacity of} \\ \text{one Track} \end{array} \times \begin{array}{l} \text{No. of Rotation} \\ \text{in one Second} \end{array} \right\}$$

Goal: To Minimize the ~~Seke~~ Seek (T).

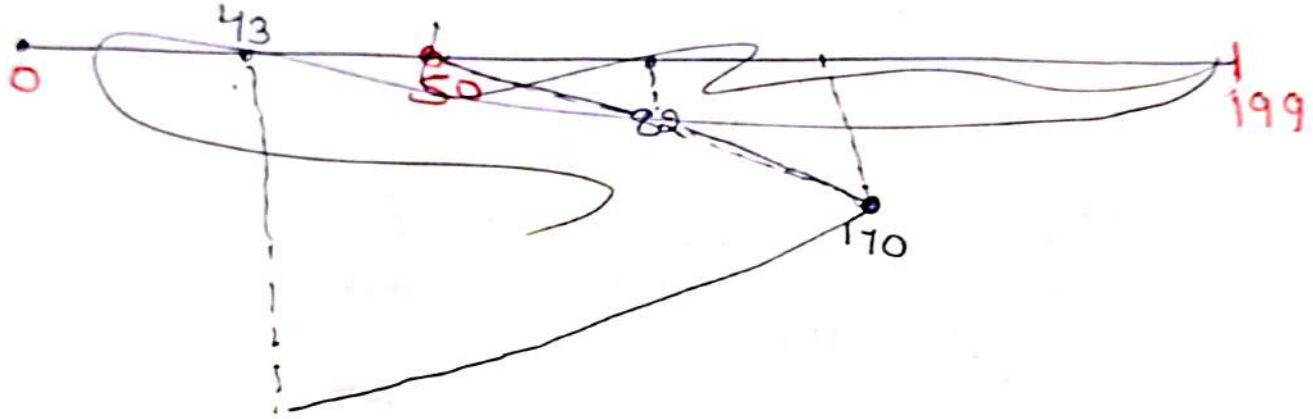
Disk Scheduling Algo

- FCFS (First come First serve).
- SSTF (Shortest Seek (T) First)
- SCAN (Elevator)
- LOOK
- CSCAN (circular SCAN)
- CLOOK (circular LOOK)

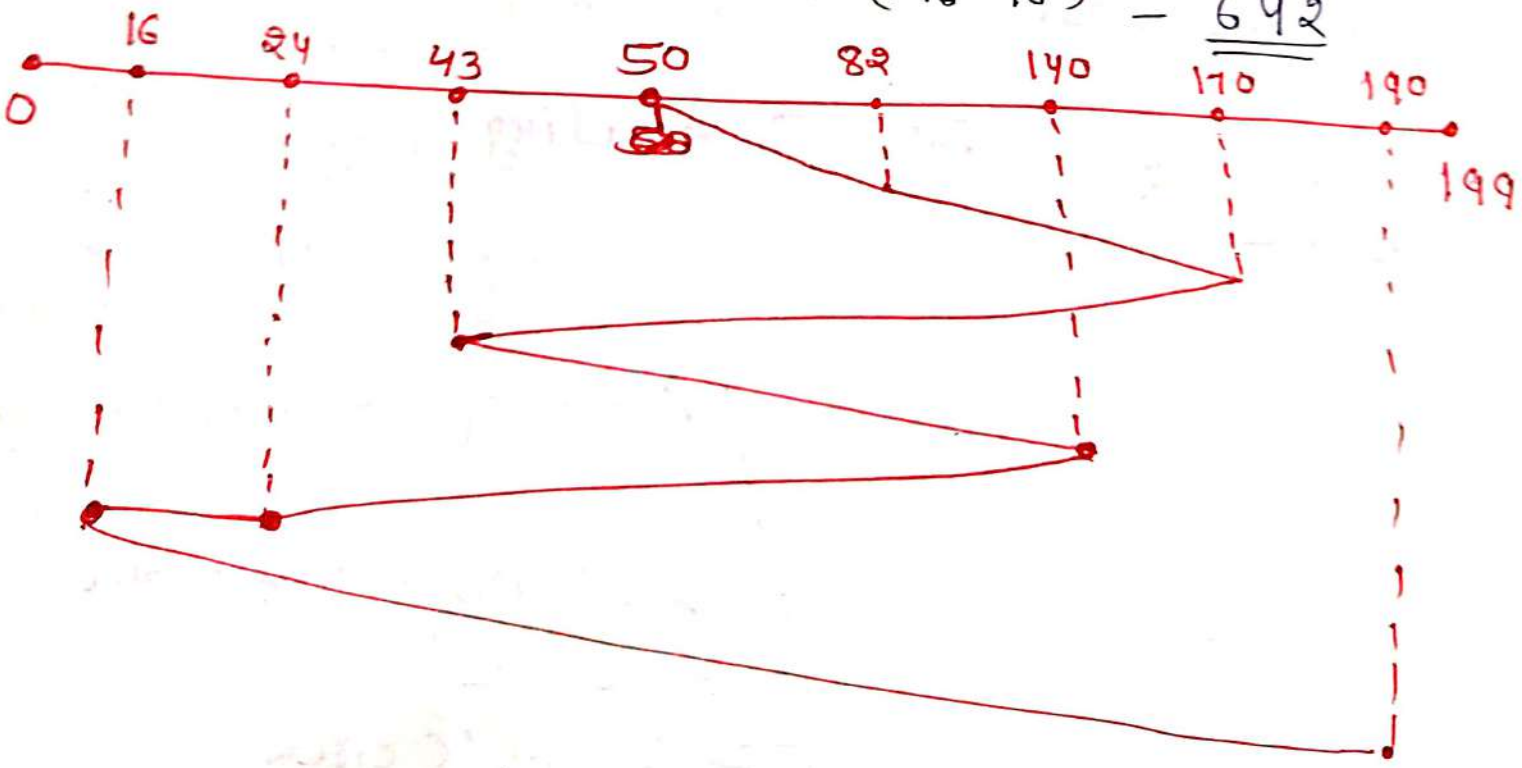
FCFS

Q1 A disk Contains 200 Tracks (0 - 199) Request Queue Contains Track No. 82, 170, 43, 140, 24, 16, 190, respectively Current Position of R/W head = 50

~~Calc~~ Caclulate total No. of Tracks Movement by R/W head.

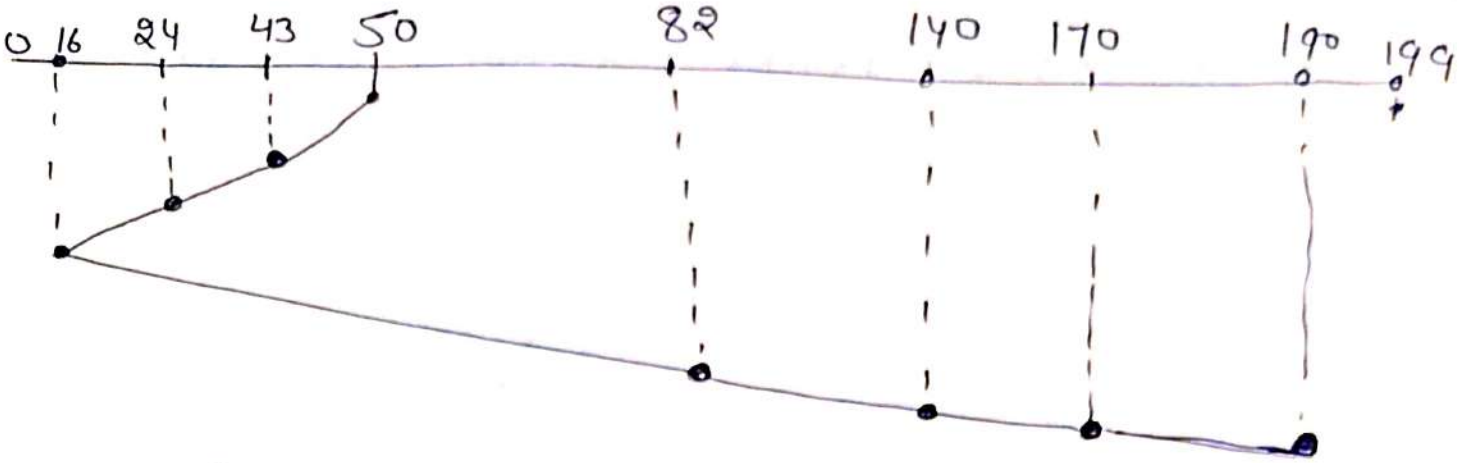


→ (170-50) + (170-43) + (140-43) + (140-16) + (190-16) = 642



SSTF (Shortest Seek Time First)

82, 170, 43, 140, 24, 16, 190

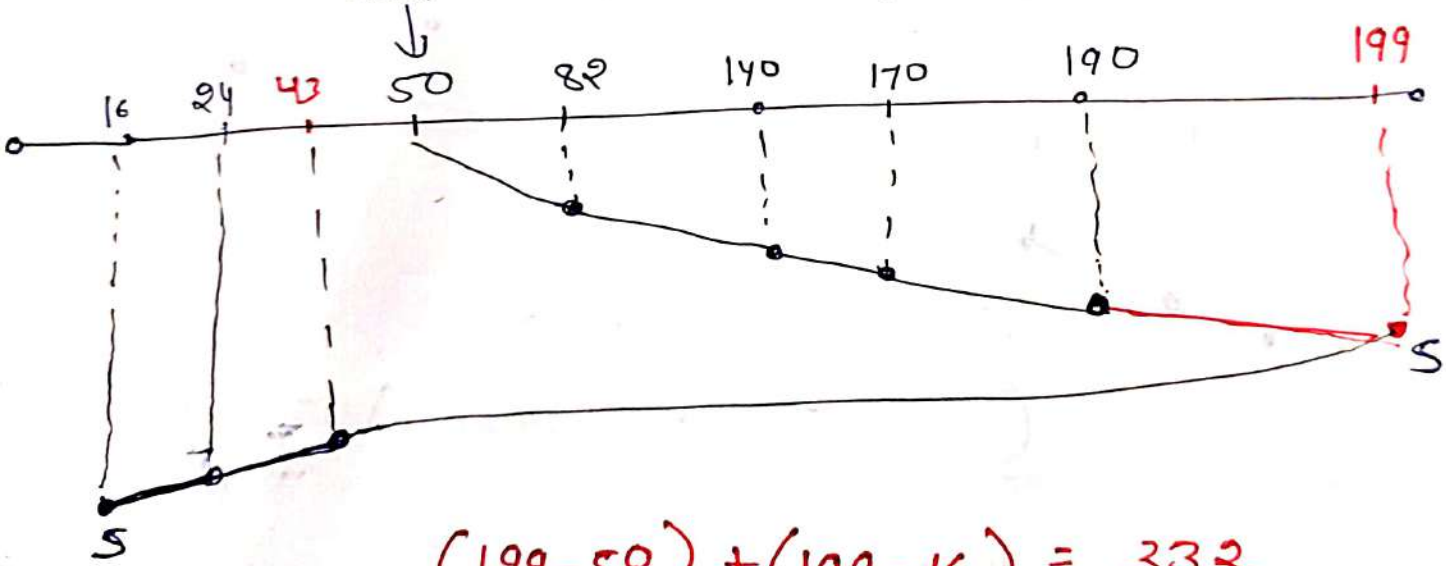


$(50-16)+(190-16) = \underline{\underline{208}}$

SCAN Algo → elevator Algo.

Direction is Towards larger value.

R/W head 82, 170, 43, 140, 24, 16, 190



$(199-50)+(199-16) = \underline{\underline{332}}$

Look Algo

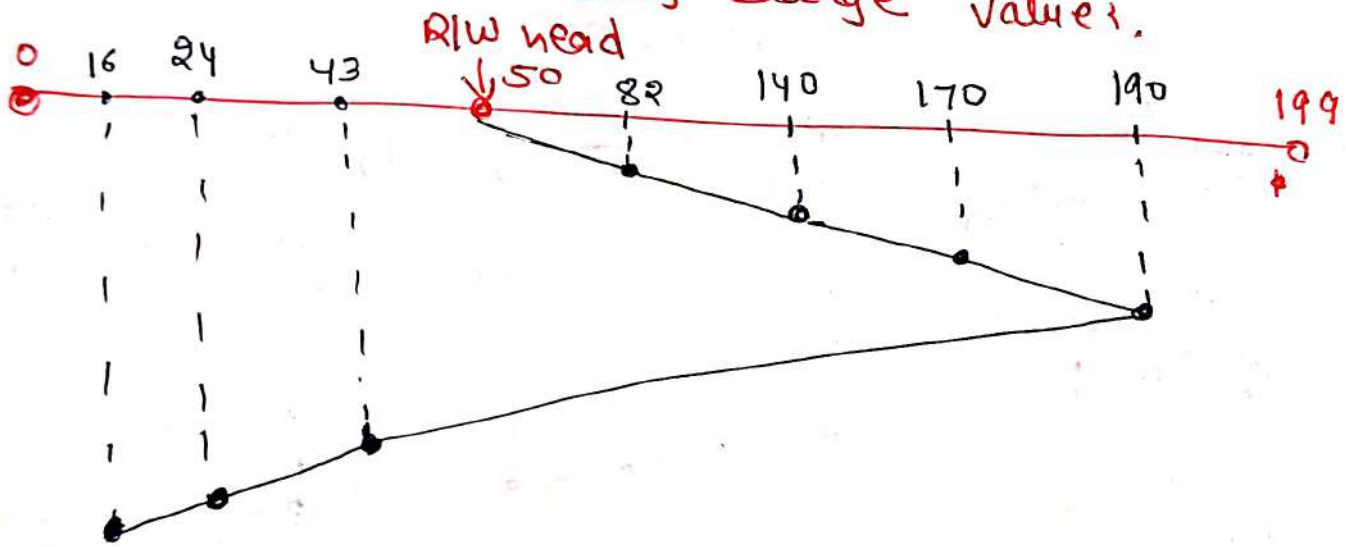
82, 170, 43, 140, 24, 16, 190.

A disk Contains 200 Tracks (0-199) Request queue Contains Track No.

- Current Position of R/W head = 50
- Caclulate total no. of Tracks Movement by R/W head using LOOK?

→ if R/W head takes 1ms to move from one from one Track to another then total (T) Taken.

* Diriction is Towards large values.



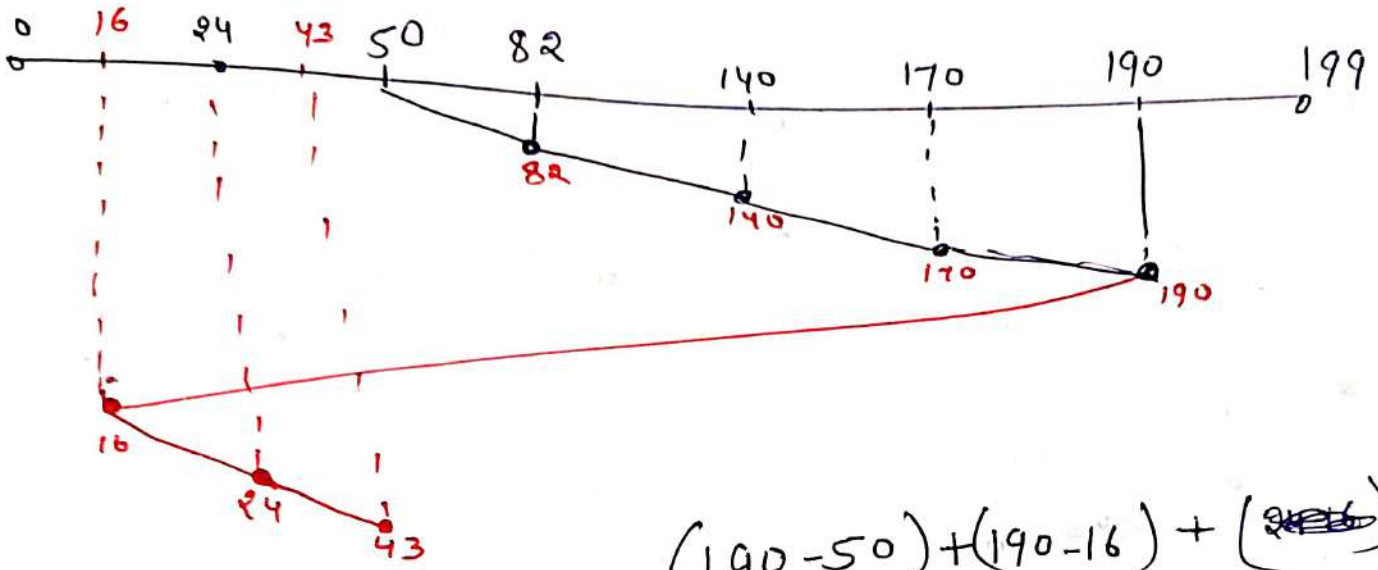
$$(190 - 50) + (190 - 16) = 314$$

$$314 \times 1ms = 314ms.$$

C-LOOK

82, 170, 43, 140, 24, 16, 190.

Direction is Towards large value.

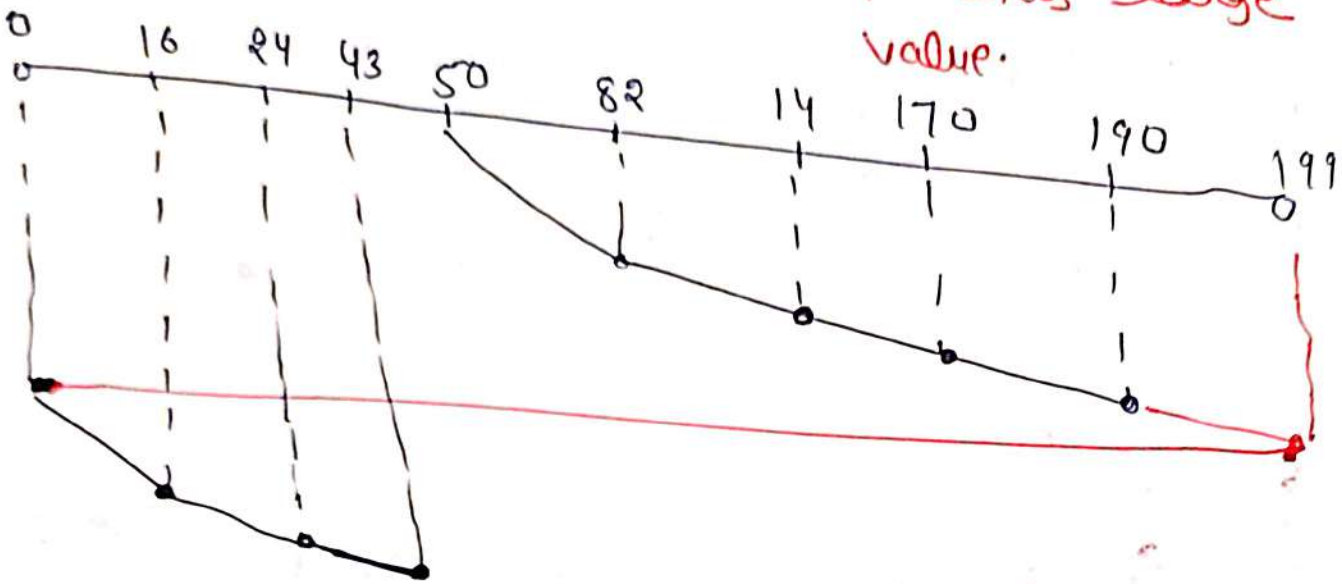


$(190-50)+(190-16)+(43-16) = 341$

C- SCAN Algo :.

82, 170, 43, 140, 24, 16, 190

→ Direction is Towards large value.



$$(199-50)+(199-0)+(43-0)$$
$$= \underline{\underline{391}}$$

Platters
↓
Tracks
↓
Sector → Three part

Disk Formatting (Partitions)

Header (Syf.	Data	ECC Error Correction Code.

→ A disk can be futher subdivided into partitions : a contiguous groups of cylinders.

→ Each partition is viewed by the (OS) as a logical disk.

Next, we'll look at the partition Table from a running linux installation.

Disk Formatting

(2) Logical Formating or high-level Formating — Partition Down by user.

(1) Physical Formating or Low level Formating.

* Boot Block :-

— Contains Beet strap loader.

Booting

The activity To start a computer is known as Booting.

Cold Booting

The process to start computer from begining is called called beeting.

Warm booting

The processing To Restart The computer.

Disk Management

- Disk Formating
- Booting disk
- bad-block

* Boot Block :-

When computer starts it needs initial program to run. It initializes

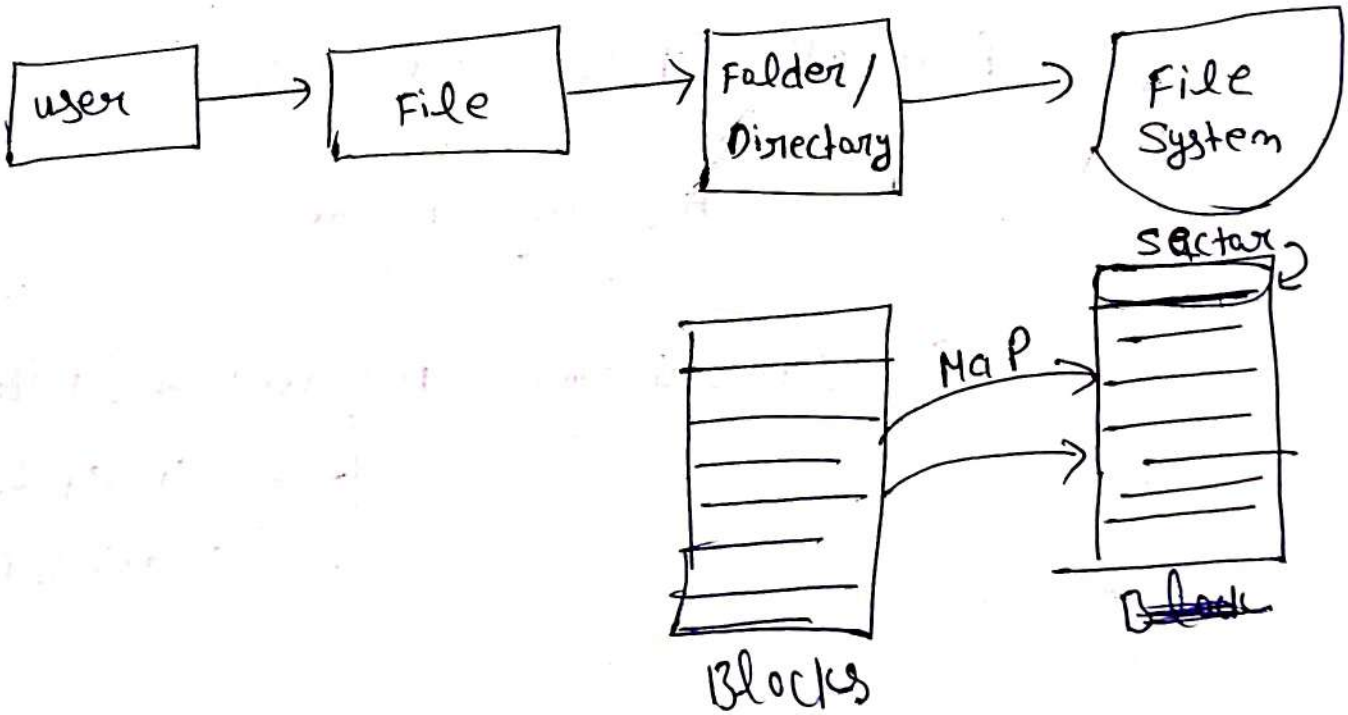
- → CPU register.
- → device controllers
- → Content of main memory
- → Start the OS.

Boot strap Program
(ROM)

→ No initialization

File system in operating system

↓ S/W
- → How stored
- → How Fetched.



Operations on Files

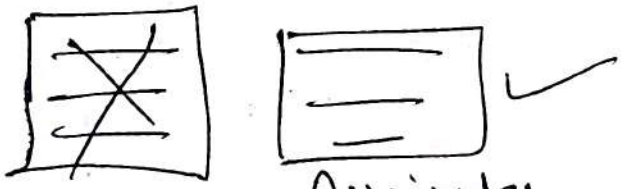
1) Creating (Touch)/CAT
2) Reading
3) Writing
4) Deleting
5) Truncate
6) Repositioning

File Attributes

1) Name
2) Extension (Type)
3) Identifier
4) Location
5) Size
6) Modified date, created date.
7) Protection/Permission
8) Encryption, Compression.

Deleting =

File Attributes

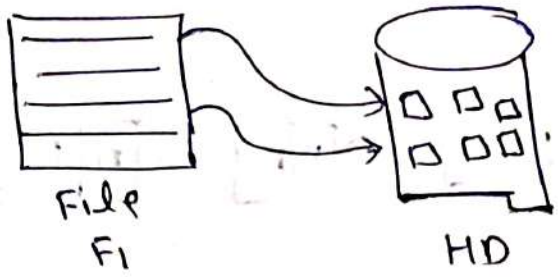
Truncate 

Attributes.

Allocation Methods in OS

Allocation Methods

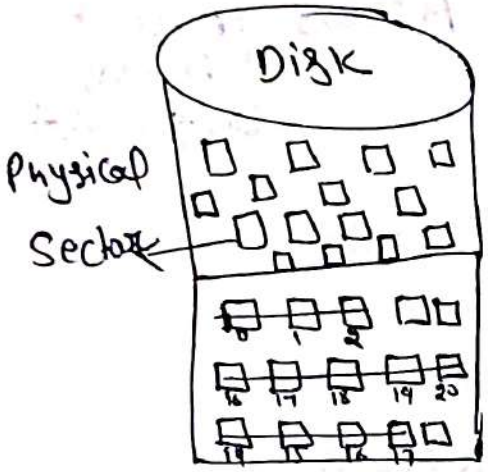
- Contiguous Allocation
- Non Contiguous Allocation
  - Linked List
  - Indexed Allocation



Achieving

1) Efficient disk utilization.
2) Access Faster.

* Contiguous Allocation



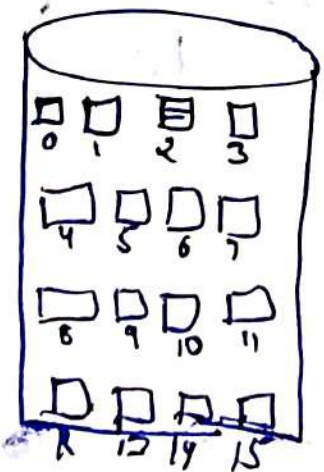
Directory

File	Start	Length
A	0	3
B	16	5
C	14	4

Advantages: - 1) Easy to implement
2) Excellent read performance

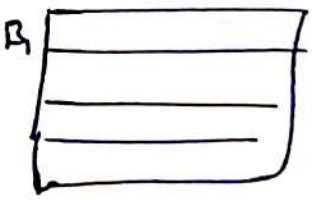
Disadvantages: - 1) Disk will become Fragmented.
2) Difficult to grow File.

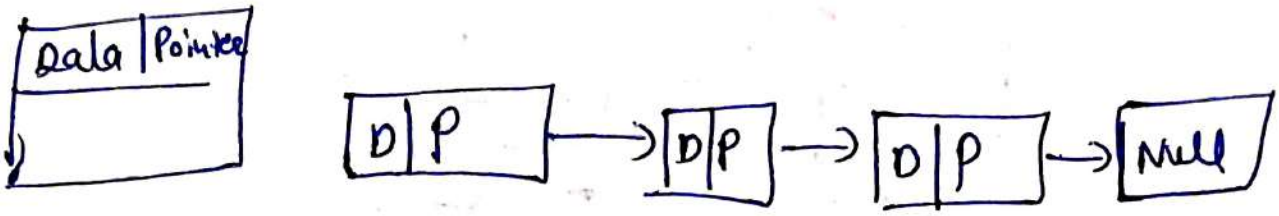
* Non Contiguous Allocation → 1) Linked List Allocation.



Directory

File	Start
A	2

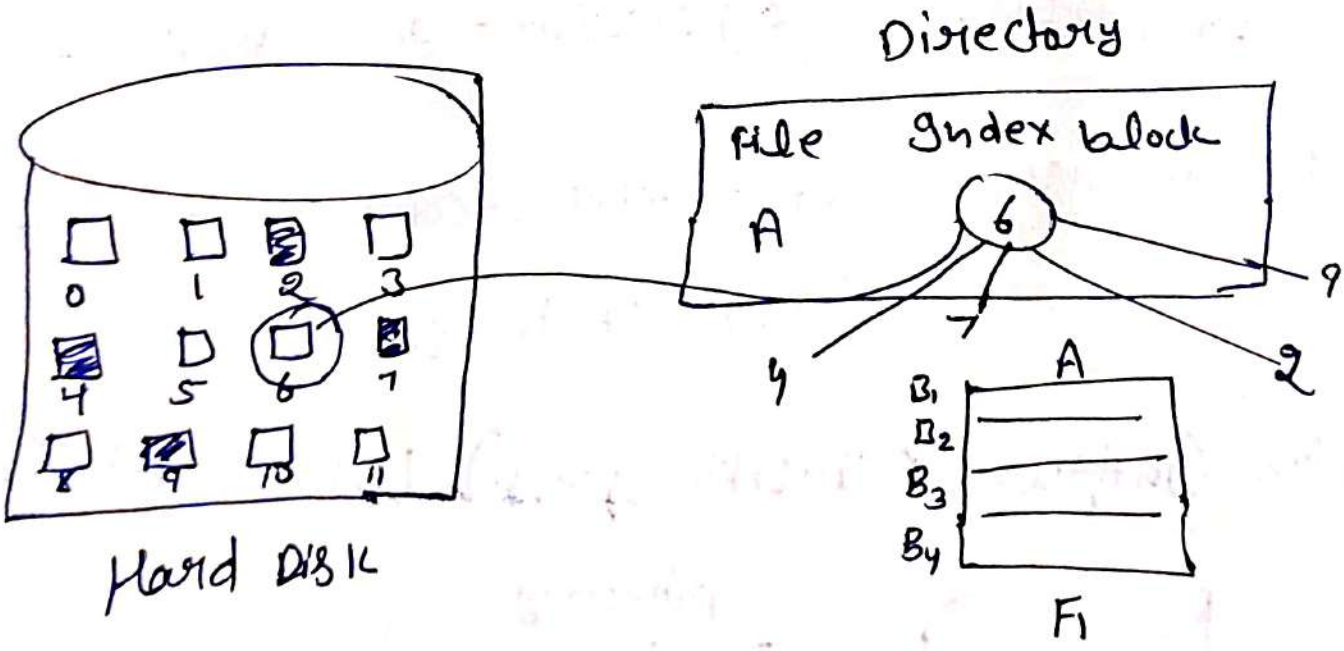




Advantages:-
1) No External Fragmentation
2) File Size Can increase

Disadvantages:-
1) Large Seek Time.
2) Random ~~Access~~ Access / Direct Access Diffiicult
3) overhead of Pointer Mems.

2) Indexed File Allocation :-



Advantages	DisAdvantages
1) support direct access.	1) Pointer overhead.
2) No External Fragmentation.	2) Multilevel Index.